

Heat, Wealth, and Vegetation: Urban Heat Island Drivers in Lincoln, Nebraska

Math in the City Spring 2026

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Abstract

On a summer afternoon, Lincoln, Nebraska, surface temperatures can vary dramatically from one neighborhood to the next, these differences are not random, but deeply tied to the city's environmental and socioeconomic landscape. This study investigates the spatial distribution and underlying drivers of the Urban Heat Island (UHI) effect by integrating environmental and demographic data at the census tract level from 2020 to 2025. A comprehensive dataset was constructed by aggregating land surface temperature (LST), vegetation density (NDVI), impervious surface extent (NDBI), and key socioeconomic indicators, including median household income, poverty rate, and renter occupancy. Exploratory data analysis reveals substantial spatial heterogeneity and abnormal distributions across variables, with multi-modal temperature patterns reflecting distinct urban typologies. Comparative analysis between the hottest and coolest census tracts shows statistically significant differences on both environmental and social dimensions. Higher temperatures are strongly associated with lower vegetation cover, greater impervious surface concentration, lower household income, and higher renter occupancy rates. These relationships are further supported by correlation analysis and regression based visualizations, consistently demonstrating that disadvantaged communities experience elevated heat exposure. The findings highlight a clear pattern of urban heat inequality in Lincoln, where vulnerable populations are disproportionately affected by extreme surface temperatures. By linking environmental conditions with social vulnerability, this study emphasizes the need for equitable urban planning interventions, such as targeted greening and heat mitigation strategies, to address the uneven impacts of climate-related heat risks.

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Chapter 1

Introduction

Urban Heat Island (UHI) effects, where urbanized areas experience significantly higher temperatures than surrounding rural regions, represent a growing environmental and public health concern worldwide. These temperature disparities arise primarily from the alteration of natural land surfaces into impervious built environments, which absorb and retain heat more effectively. The consequences of UHI are far-reaching, including increased energy consumption, elevated air pollution levels, and heightened health risks such as heat stress and heat-related mortality. Critically, the impacts of UHI are not evenly distributed across urban populations, often exacerbating existing social and environmental inequities. This study focuses on Lincoln, Nebraska, to investigate how physical landscape characteristics and socioeconomic variables interact to influence local heat exposure patterns over a longitudinal period from 2020 to 2025.

The core environmental variables examined in this study are the Land Surface Temperature (LST), Normalized Difference Vegetation Index (NDVI), and Normalized Difference Built-up Index (NDBI). LST is a direct measure of the heat emitted from the Earth's surface, derived from satellite thermal imagery, and serves as a proxy for surface-level heat exposure experienced by urban residents. NDVI quantifies vegetation density by measuring the difference between near-infrared and red light reflectance, with higher values indicating more abundant and healthier vegetation. Because vegetation provides shading and enhances evapotranspiration, it is hypothesized that higher NDVI values correlate with lower LST, thus mitigating the UHI effect. Conversely, NDBI captures the extent of impervious, built-up surfaces such as concrete and asphalt, which typically absorb and store heat, leading to elevated LST. Therefore, a positive association between NDBI and LST is expected, indicating that highly urbanized areas experience greater heat exposure.

Beyond physical landscape factors, socioeconomic variables play a crucial role in shaping vulnerability to urban heat. Metrics such as poverty rate, median household income, renter occupancy percentage, percentage of elderly residents, disability prevalence, racial composition, and limited English proficiency are integrated to assess social dimensions of heat exposure. Lower-income and renter-occupied neighborhoods often have less access to cooling infrastructure and mature vegetation, increasing residents' heat risk. Similarly, elderly populations and individuals with disabilities are more physiologically vulnerable to heat stress, making the demographic composition of neighborhoods a critical factor in understanding heat-related health disparities. Non-white populations and households with limited English proficiency may face additional barriers to resources and informa-

tion, compounding their vulnerability.

This study hypothesizes that census tracts with higher poverty rates, greater renter occupancy, larger elderly and disabled populations, and higher percentages of non-white or limited English proficiency households will exhibit higher average LST values, reflecting disproportionate heat burdens. By combining these socioeconomic indicators with environmental indices, the research aims to provide a nuanced understanding of how heat exposure intersects with social vulnerability in Lincoln. This integrated approach is essential for identifying priority areas for intervention and informing equitable urban planning and climate resilience strategies.

Through this comprehensive analysis, the project not only advances the scientific understanding of UHI dynamics in a mid-sized American city but also equips local policymakers with actionable insights. By revealing spatial patterns of heat exposure and their social determinants, the study supports targeted investments in urban greening, cooling infrastructure, and community outreach to protect the city’s most vulnerable residents from the escalating risks of extreme heat.

1.1 Motivation

Rooted in the need to address the “luxury effect,” where urban heat exposure forms unequal burdens to low-income and renter-occupied neighborhoods, creates significant health disparities. [7, 1] Preliminary data indicates that while Lincoln’s 2021 Climate Action Plan [2] has brought attention to urban cooling, the city still experiences a pronounced UHI effect, with surface temperatures consistently higher in densely built environments compared to greener, wealthier areas. [3] By analyzing a 5-year longitudinal dataset (2020–2025), this study provides the evidence needed to transition from general climate awareness to targeted, data-driven interventions. The project serves as a critical tool for local policymakers and public health officials, offering a granular map of heat vulnerability that can inform equitable investments in urban forestry, public cooling infrastructure, and disaster resilience planning to protect Lincoln’s most at risk residents.

1.2 Overview of the Project

The Urban Heat Island effect, characterized by urban areas experiencing extreme temperatures compared to their rural surroundings, is an increasing public health and environmental justice crisis. National assessments across the 175 largest urbanized areas in the US reveal that the intensity of surface urban heat island (SUHI) is disproportionately distributed, with people of color and those living below the poverty line exposed to a higher intensity of heat in 97% of the cities studied [4]. Within Nebraska, heat vulnerability is recognized as an issue where socioeconomic drivers vary by urbanization scale; high-vulnerability clusters appear most prominently in metropolitan areas such as Omaha and Lincoln [3]. This study seeks to investigate these disparities at a granular level. By establishing a longitudinal data set that spans 2020-2025, the objective is to address how environmental and socioeconomic factors influence the spatial distribution of UHIs in Lincoln, Nebraska, defining local observations within broader national and state-level trends of thermal inequity.

Existing literature identifies land cover, specifically vegetation density and proximity to blue spaces, as a primary determinant of a neighborhood’s thermal health. Research identifies that increased urban vegetation, quantified through the NDVI, provides a critical cooling effect that mitigates land surface temperature (LST) [9]. The presence of water bodies such as lakes and wetlands offers localized thermal regulation. This study aims to quantify these relationships across Lincoln’s census block groups by analyzing the correlation between NDVI, proximity to water bodies, and derived satellite temperature data. This environmental focus allows for a precise evaluation of how Lincoln’s natural and built infrastructure influences local temperature patterns, addressing whether the city’s green and blue spaces are equitably distributed to provide maximum thermal relief.

Socially, the distribution of heat exposure is trended towards neighborhood demographics and housing tenure. National and local studies suggest a “luxury effect,” where high-income, owner-occupied neighborhoods benefit from mature tree canopies and lower impervious surface cover, while lower-income areas and high-renter-occupancy zones often suffer a “heat penalty” due to a lack of cooling infrastructure [6] & [5]. By utilizing Lincoln Vital Signs income maps and renter occupancy data with time-series temperature model, this study explores whether lower median household incomes and higher renter percentages directly correlate with elevated urban heat exposure. This targeted approach moves beyond general context to test if Lincoln’s specific neighborhood demographics dictate the lived experience of extreme heat, informing climate resilient policies tailored to the city’s most vulnerable populations.

1.3 Organization of the Report

This report flows from broad motivation and background of the Urban Heat Island (UHI) effect to a detailed, data-driven analysis of its drivers in Lincoln, Nebraska.

- **Chapter 1:** Introduces the problem of urban heat inequality, outlines the key environmental and socioeconomic variables, and presents the central research questions and hypotheses. It also provides motivation and situates the project within existing literature on UHI and environmental justice.
- **Chapter 2:** Describes the data sources and assumptions underlying the analysis. Details how environmental and socioeconomic datasets are collected, processed, and integrated at the census tract level, along with the rationale for variable selection and seasonal filtering.
- **Chapter 3:** Provides the mathematical and statistical background necessary for the study. This includes definitions and formulations for exploratory data analysis, correlation measures, hypothesis testing, spatial autocorrelation (LISA), and the machine learning framework used in later sections.
- **Chapter 4:** The core analysis. It applies the methods described earlier to examine relationships between land surface temperature and environmental and socioeconomic factors. This chapter includes summary statistics, correlation analysis, hotspot identification, spatial visualization, and modeling results.

- **Chapter 5:** Summarizes the key findings, interprets the results in the context of urban heat inequality, and discusses implications for urban planning and policy. Reflections on the limitations of the study and potential directions for future research are discussed.

Chapter 2

Data and Assumptions

2.1 Data

This study integrates multiple datasets to comprehensively analyze the Urban Heat Island (UHI) effect in Lincoln, Nebraska, focusing on environmental and social inequities from 2020 to 2025.

Satellite Thermal Imagery: We utilize Landsat 8 and 9 Collection 2 Level 2 data accessed via the Google Earth Engine API. Data are filtered to include only summer months (June through August) with less than 10% cloud cover. This seasonal filtering is critical because summer represents the period of highest heat exposure and thus the greatest health risk from UHI effects. Selecting cloud-free images ensures data quality and reliability in Land Surface Temperature (LST) measurements.

Land Cover Data: The National Land Cover Database (NLCD) provides detailed classification of developed and natural land types, allowing us to characterize urban infrastructure and natural features that influence local microclimates.

Vegetation and Built-up Indices: NDVI (Normalized Difference Vegetation Index) and NDBI (Normalized Difference Built-up Index) are calculated annually from satellite imagery. These indices serve as proxies for vegetation density and impervious surfaces, respectively, key physical drivers of heat variation in urban areas.

Socioeconomic Data: Socioeconomic variables were chosen through a literature review [3]. Specifically, we chose three socioeconomic variables to thoroughly analyze: poverty rate percentage, median household income, and renter-occupied percentage. Additionally, we had the following variables that were also used throughout the study, but weren't thoroughly studied: Percent Age 65+, Count Age 65+, Elderly Living Alone Percent, Count of Elderly Living Alone, Percent of Persons with Disability, Count of Disabled Persons, Non-White Percent, Count of Non-White Persons, Limited English Households Percent, and Count of Limited English Households Percent. We incorporate American Community Survey (ACS) 5-year estimates at the census tract level, capturing variables such as poverty rate, disability prevalence, racial composition, elderly living alone, limited English proficiency, and renter occupancy. These metrics provide insight into social vulnerability and potential inequities in heat exposure.

Data Processing: Using the Python `rasterstats` module, raster data (LST, NDVI, NDBI) are aggregated by census tract polygons through zonal statistics, producing a master dataset with averaged environmental and socioeconomic variables per tract per year. This spatial aggregation aligns environmental and demographic data at a neighborhood scale appropriate for urban planning analysis.

We retain variables that directly relate to environmental conditions or social vulnerability to heat, ensuring that the analysis targets the intersection of physical and demographic drivers of UHI exposure.

2.2 Assumptions

- **Spatial Units:** Census tracts are assumed to be appropriate units for capturing neighborhood-level variations in heat exposure and socioeconomic characteristics, balancing spatial resolution and data availability.
- **Index Validity:** NDVI and NDBI are accepted as reliable proxies for vegetation density and impervious surface extent, respectively, consistent with established urban heat research.
- **Temporal Relevance:** Averaging LST over summer months captures meaningful heat exposure relevant to public health outcomes, reflecting peak UHI conditions.
- **Socioeconomic Data Representativeness:** ACS 5-year estimates sufficiently represent tract populations during the study period, despite inherent temporal averaging.
- **Modeling Approach:** Initial relationships between heat exposure and predictors can be modeled linearly or through factor analysis before applying more complex machine learning techniques.
- **Spatial Autocorrelation:** Spatial dependence is expected and will be addressed using Local Indicators of Spatial Association (LISA) cluster analysis to identify statistically significant spatial patterns.

These assumptions are grounded in the spatial and temporal resolution of the data and the established validity of chosen indices in urban heat studies.

Chapter 3

Mathematical Background

3.1 Exploratory Data Analysis

We will begin by computing summary statistics (mean, median, standard deviation, min, and max) for all key variables including LST, NDVI, NDBI, median household income, poverty rate, and renter occupancy. These provide essential insights into central tendencies and spread, helping identify patterns or anomalies in heat exposure and socioeconomic conditions. For example, the mean and median reveal typical values, while standard deviation highlights variability across neighborhoods. Histograms will be generated for each to assess the shape of their distributions and identify any remaining outliers or skewness. This step establishes a baseline understanding of how heat, vegetation, and socioeconomic conditions are distributed across Lincoln’s census tracts before any relational analysis is conducted. We compute summary statistics for each variable X (e.g., LST, NDVI, income):

$$\text{Mean}(X) = \frac{1}{n} \sum_{i=1}^n X_i,$$

$$\text{Median}(X) = \text{middle value of ordered } \{X_i\},$$

$$\text{Standard Deviation}(X) = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2}$$

Minimum and maximum values are:

$$\min(X) = \min_i X_i, \quad \max(X) = \max_i X_i$$

Histograms visualize the empirical distribution $f_X(x)$, revealing skewness, outliers, and multimodal patterns, informing subsequent modeling choices and data transformations if necessary. The initial analysis involves computing summary statistics—mean, median, standard deviation, minimum, and maximum—for all key variables, including LST, NDVI, NDBI, median household income, poverty rate, and renter occupancy. This quantitative overview establishes baseline distributions and variability within Lincoln’s census tracts.

This foundational step sets the stage for deeper relational analyses, such as spatial autocorrelation and machine learning, by ensuring a clear understanding of the data’s structure and quality.

3.2 Correlation Analysis

We will compute Pearson, Spearman rank, Kendall’s Tau, and Distance correlations between LST and each predictor variable, including NDVI, NDBI, median household income, poverty rate, and renter occupancy percentage. Running each method allows us to detect nonlinear relationships that Pearson alone would miss. This step tests the core hypotheses — that higher LST is associated with lower vegetation, lower income, and higher renter density — and determines which relationships are strong enough to warrant deeper modeling. In summary, using all four methods provides:

- An understanding of the linear component (Pearson).
- General directional trends (Spearman, Kendall).
- Detection of any underlying dependency, including complex non-linear ones (Distance Correlation).

This approach ensures that we do not miss important relationships due to assumptions about linearity or distribution, offering a better understanding of the data.

We calculate correlation coefficients between LST and each predictor X :

- **Pearson correlation** (measures linear association):

$$r = \frac{\sum_{i=1}^n (LST_i - \overline{LST})(X_i - \bar{X})}{\sqrt{\sum_{i=1}^n (LST_i - \overline{LST})^2} \sqrt{\sum_{i=1}^n (X_i - \bar{X})^2}}$$

- **Spearman rank correlation** ρ (measures monotonic association using ranks R):

$$\rho = 1 - \frac{6 \sum_{i=1}^n (R_{LST_i} - R_{X_i})^2}{n(n^2 - 1)}$$

- **Kendall’s Tau** (measures monotonic association using concordant/discordant pairs):

$$\tau = \frac{C - D}{\binom{n}{2}}$$

- C = number of concordant pairs
- D = number of discordant pairs
- $\binom{n}{2} = \frac{n(n-1)}{2}$ is the total number of pairs

$$\tau = \frac{2(C - D)}{n(n - 1)}$$

- **Distance Correlation** (measures any dependence, not just linear/monotonic and will always be nonnegative):

$$\text{dCor}(X, Y) = \frac{\text{dCov}(X, Y)}{\sqrt{\text{dVar}(X) \text{dVar}(Y)}}$$

$$\text{dCov}^2(X, Y) = \frac{1}{n^2} \sum_{i=1}^n \sum_{j=1}^n A_{ij} B_{ij}$$

– A_{ij}, B_{ij} are double-centered distance matrices

$$\text{dVar}(X) = \text{dCov}(X, X), \quad \text{dVar}(Y) = \text{dCov}(Y, Y)$$

These test hypotheses such as:

Higher $LST \leftrightarrow$ Lower $NDVI$, Lower income, Higher renter occupancy

To elaborate a bit on the significance of each correlation, we have provided a summary of each of them in plain English, which helps unpack their purpose and the meaning behind the respective formulas:

- **Pearson Correlation:** This measures the strength and direction of a linear relationship between two variables. A value of +1 indicates a perfect positive linear relationship (as one variable increases, the other increases proportionally), -1 indicates a perfect negative linear relationship (as one increases, the other decreases proportionally), and 0 indicates no linear relationship.
- **Spearman Correlation:** This measures the strength and direction of a monotonic relationship between two variables, meaning it assesses how well the relationship between two variables can be described using a monotonic function (either consistently increasing or consistently decreasing). It's based on the ranks of the data values rather than the actual values themselves, making it less sensitive to outliers and non-linear but monotonic relationships.
- **Kendall's Tau:** A non-parametric measure of the strength and direction of association between two variables. Like Spearman's rank correlation, it assesses monotonic relationships (where variables tend to move in the same or opposite direction, but not necessarily at a constant rate). However, it does so by counting concordant and discordant pairs, making it particularly robust to outliers and non-normally distributed data.
- **Distance Correlation:** A robust, powerful and relatively new measure of dependence between two random variables. Unlike Pearson correlation, which only detects linear relationships, and Spearman/Kendall, which detect monotonic relationships, Distance Correlation is capable of detecting any type of statistical dependence, including non-linear and non-monotonic relationships. If the distance correlation is zero, it implies that the variables are statistically independent. If it's 1, it implies perfect dependence. In general, a moderate Distance Correlation indicates a significant dependency because a value of zero strictly implies statistical independence. Therefore, any non-zero value, even in the moderate range, signifies a real, non-random relationship between variables.

3.3 Heat Hotspot Identification

We rank the census tracts by mean summer land surface temperature. The ten hottest and ten coolest tracts will be extracted and compared across environmental and socioeconomic variables in a summary table. This provides an interpretable, concrete illustration of whether Lincoln’s hottest neighborhoods systematically differ from its coolest ones in terms of income, vegetation cover, and renter occupancy.

Census tracts are ranked by mean summer LST:

$$\text{Rank}(i) \text{ based on } \overline{LST}_i, \quad i = 1, \dots, n$$

The top 10 hottest and bottom 10 coolest tracts are compared using Welch’s t-test:

$$t = \frac{\bar{X}_{hot} - \bar{X}_{cool}}{\sqrt{\frac{s_{hot}^2}{n_{hot}} + \frac{s_{cool}^2}{n_{cool}}}}$$

where $\bar{X} = \frac{\sum_{i=1}^n (x_i)}{n}$ are the mean of each category, $s^2 = \frac{\sum_{i=1}^n (x_i - \bar{X})^2}{n-1}$ are the sample variances of each category, and $n = 10$ is the sample size.

The t-value that we get from Welch’s t-test tells us how significant the difference between the hottest and coolest census tracts are for each category. A positive value tells us that the category’s value is higher in the hottest tracts, while a negative value tells us that the category’s value is higher in the coolest tracts. If the absolute value of a given t-value is greater than 2.1, this tells us that the category is statistically significant, which means that it has a strong relationship with the Land Surface Temperature.

Looking at [5], we can see that working with census tract data to compare different variables is a proven method that can reveal some interesting correlations and connections between land surface temperature and other census tract variables. We can also hypothesize that we will see high t-values for variables relating to income and poverty.

3.4 Data Integration and Visualization

Scatter plots will be produced for key variable pairs (LST vs. NDVI, LST vs. income, LST vs. poverty rate, LST vs. renter occupancy), with points color-coded by a third variable such as poverty rate to reveal compounding relationships. A correlation heatmap will summarize pairwise relationships across all variables simultaneously. These visualizations will accompany the correlation analysis in the final report.

Mapping spatial distributions of LST, NDVI, income, and renter occupancy to visually identify potential heat hotspots and vulnerable populations.

NDVI: $\frac{\text{Band5(NIR)} - \text{Band4(Red)}}{\text{Band5(NIR)} + \text{Band4(Red)}}$ Where, 0.6 to 0.9: Dense vegetation (e.g., forests); 0.2 to 0.5: Sparse vegetation (e.g., grasslands); -0.1 to 0.1: desolated areas (e.g., impenetrable surfaces/urban); Negative: Water, snow, or clouds.

NDBI: $\frac{\text{Band6(SWIR)} - \text{Band5(NIR)}}{\text{Band6(SWIR)} + \text{Band5(NIR)}}$

Scatter plots illustrate relationships between pairs (LST, X). Points are color-coded by a third variable Z (e.g., poverty rate) to reveal compounding effects.

3.5 Spatial Mapping and Clustering

3.5.1 Global Spatial Autocorrelation

To formally assess whether LST exhibits spatial structure, we computed Global Moran's I, a measure of spatial autocorrelation. Moran's I evaluates whether similar values are clustered together across the study region.

$$I = \frac{n}{S_0} \cdot \frac{\sum_{i=1}^n \sum_{j=1}^n w_{ij} (LST_i - \overline{LST})(LST_j - \overline{LST})}{\sum_{i=1}^n (LST_i - \overline{LST})^2}$$

where n is the number of spatial units, w_{ij} represents the spatial weight between tracts i and j , $S_0 = \sum_i \sum_j w_{ij}$, and $\overline{LST}$ is the mean land surface temperature.

A positive Moran's I indicates clustering of similar values (high with high, low with low), while a negative value indicates dispersion. Statistical significance is evaluated using permutation-based inference.

3.5.2 Local Spatial Autocorrelation (LISA)

To identify localized clustering patterns, we computed Local Indicators of Spatial Association (LISA), specifically Local Moran's I, for each census tract i :

$$I_i = \frac{(LST_i - \overline{LST})}{m_2} \sum_{j=1}^n w_{ij} (LST_j - \overline{LST})$$

where

$$m_2 = \frac{1}{n} \sum_{k=1}^n (LST_k - \overline{LST})^2$$

and w_{ij} represents the spatial weight between neighboring tracts.

Local Moran's I measures the similarity between a tract and its neighbors. A statistically significant positive value indicates that a tract is surrounded by tracts with similar values (high-high or low-low clustering), while a significant negative value indicates a spatial outlier (high-low or low-high).

In this analysis, spatial weights were defined using a contiguity-based approach (Queen adjacency), where neighboring tracts share a boundary or vertex. This allows for realistic modeling of spatial relationships in irregular geographic units.

3.5.3 Geographically Weighted Regression (GWR)

To model spatially varying relationships between LST and explanatory variables, we implemented a Geographically Weighted Regression (GWR) model. Unlike traditional global regression, which assumes constant relationships across space, GWR allows regression coefficients to vary by location.

The GWR model is defined as:

$$LST_i = \beta_0(u_i, v_i) + \sum_{k=1}^p \beta_k(u_i, v_i) X_{ik} + \epsilon_i$$

where (u_i, v_i) are the spatial coordinates of tract i , $\beta_k(u_i, v_i)$ are location-specific coefficients, X_{ik} are explanatory variables (e.g., NDVI, NDBI, income), and ϵ_i is the error term.

Weights are assigned using a spatial kernel function, with nearby observations receiving greater influence. In this study, an adaptive bisquare kernel was used, with bandwidth selected through cross-validation to optimize model fit.

GWR enables the detection of spatial heterogeneity, allowing us to identify where environmental and socioeconomic variables have stronger or weaker associations with temperature. Model performance is evaluated using metrics such as local R^2 , Akaike Information Criterion (AIC), and residual diagnostics.

3.6 Machine Learning and Explainability

A Random Forest regression model is trained to predict LST from the combined set of environmental and socioeconomic predictors. Random Forest is an ensemble method that builds T independent decision trees, each trained on a bootstrap sample of the data. At every split, only a random subset of $\sqrt{p}$ features is considered, which decorrelates the trees and prevents any single dominant predictor from driving every branch. The final prediction is the average across all trees:

$$L\hat{S}T_i = \frac{1}{T} \sum_{t=1}^T h_t(\mathbf{X}_i)$$

where h_t is the t -th decision tree. This ensemble approach is preferred over linear regression here because the relationships between LST and predictors such as income or renter occupancy may not be strictly linear or additive.

Model hyperparameters were set as follows. The number of trees $T = 800$ ensures ensemble variance has stabilized. The number of features considered at each split is $\lfloor \sqrt{p} \rfloor$, the standard choice for regression forests, which ensures sufficient decorrelation between trees. A minimum of 2 samples per leaf was enforced to prevent individual trees from memorizing single observations.

Given the small sample size of 78 census tracts aggregated across years, Leave-One-Out Cross Validation (LOOCV) is used in place of a standard train/test split. Holding back a fixed test set would sacrifice too large a fraction of observations for training. LOOCV maximizes training data while still producing out-of-sample predictions for every observation:

$$\text{For each } i, \quad f_{-i} \text{ is trained on all observations except } i, \quad L\hat{S}T_i = f_{-i}(\mathbf{X}_i)$$

SHAP (SHapley Additive Explanations) values are then computed on the full-data model to quantify each feature’s contribution to each prediction:

$$L\hat{S}T_i = \phi_0 + \sum_{j=1}^p \phi_j$$

where ϕ_0 is the baseline (mean) prediction and ϕ_j is the additive contribution of feature j for observation i . SHAP values identify not only which features drive LST but in which direction and by how much, providing more interpretability.

Chapter 4

Results

4.1 Exploratory Data Analysis

Table 4.1: Summary Statistics

Variable	Mean	Median	Std. Dev.	Min	Max
LST_C	37.848	38.283	2.35	32.696	44.233
NDVI	0.58	0.589	0.109	0.198	0.792
NDBI	-0.227	-0.24	0.056	-0.337	-0.031
Median Household Income	68631.872	64325.5	27559.304	20357	133506
Poverty Rate Percent	13.285	7.96	13.119	0.76	61.64
Renter Occupied Pct	41.267	35.315	25.718	2.53	100

The summary statistics indicate a non-normal distribution across most environmental and socioeconomic variables, reflecting the complex spatial heterogeneity within Lincoln’s urban fabric. Land Surface Temperature (LST) data show a high frequency of surface temperatures clustering near 38°C during summer months, highlighting persistent heat exposure in many neighborhoods. The distribution of LST is further characterized by multimodality, suggesting distinct temperature regimes that correspond to different urban typologies. Similarly, socioeconomic variables such as poverty rate and median household income exhibit right-skewed distributions, indicating that environmental hazards like heat disproportionately affect lower-income populations. The multimodal patterns observed in both housing tenure and temperature data point toward a fragmented urban landscape, where specific census tracts consistently bear a heavier burden of the Urban Heat Island effect.

Vegetation density, measured by the Normalized Difference Vegetation Index (NDVI), exhibits a skewed distribution with many areas showing low to moderate vegetation cover and fewer tracts with dense greenery, consistent with uneven urban greening. The Normalized Difference Built-up Index (NDBI) is concentrated in higher values within densely developed tracts, reinforcing the link between impervious surface extent and elevated heat exposure. Socioeconomic indicators such as renter occupancy, elderly population percentage, and disability prevalence also display spatial clustering, with vulnerable populations concentrated in hotter, less vegetated neighborhoods. The combination of these descriptive statistics underscores the intersection of environmental and social inequities

in Lincoln, laying the groundwork for subsequent spatial and machine learning analyses to identify and address heat vulnerability hotspots.

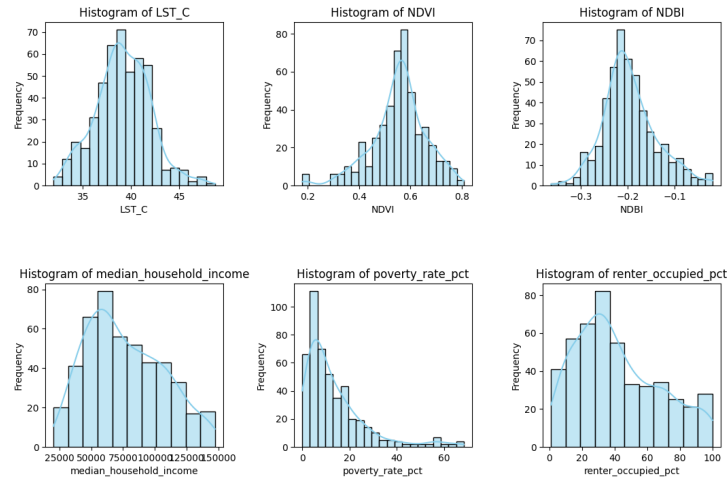


Figure 4.1: These histograms demonstrate the shape of the distributions of the focused variables.

The data reveals a stark contrast in environmental conditions. The NDVI suggests a generally green landscape, as 0.6-0.9 is interpreted as having dense, healthy vegetation. However, the LST_C histogram shows a concerning concentration of surface temperatures near 39°C, particularly in areas that may correlate with the high-poverty and high-renter density outliers seen in the socioeconomic distributions. Our socioeconomic variables show that the majority of Lincoln is clustered around \$50,000-80,000 median household income, 0-15% poverty, and 25-40% renter occupied.

We can also see how this trend in median household income has changed over the years by referring to the Household Income by Census Tract Map produced by Lincoln Vital Signs [8].

4.2 Correlation Analysis

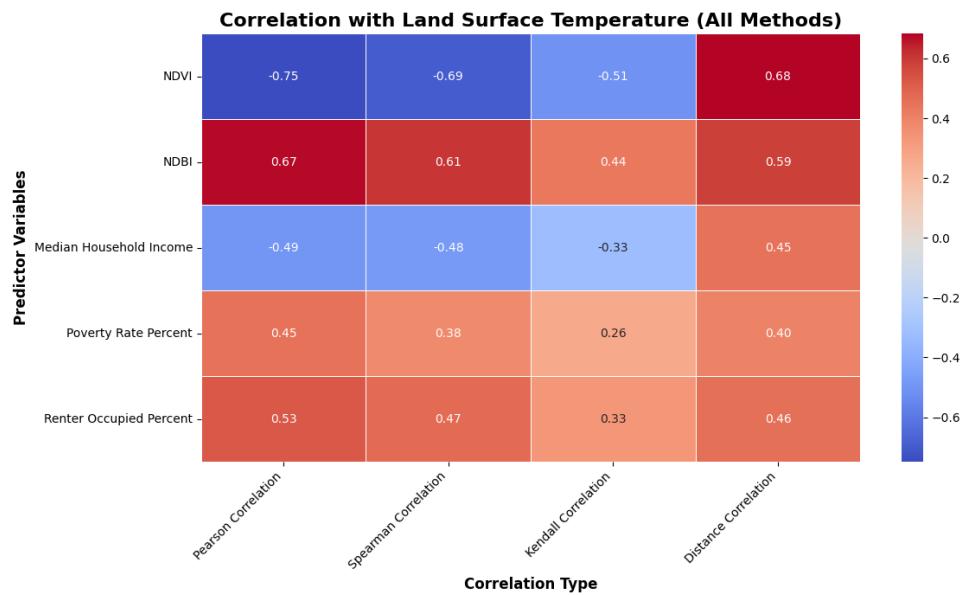


Figure 4.2: Pearson and Spearman Correlation Data

Interpretation of Pearson and Spearman Correlations

Looking at the correlation and the heatmap, the data suggests the following about the relationship between Land Surface Temperature and predictor variables:

1. NDVI (Normalized Difference Vegetation Index)

- Pearson: -0.75
- Spearman: -0.69
- Interpretation: There is a strong negative correlation between NDVI and Land Surface Temperature. This means that as vegetation density (NDVI) increases, the Land Surface Temperature tends to decrease. This is consistent with the cooling effect of vegetation, as it holds moisture (which strongly aligns with the hypothesis).

2. NDBI (Normalized Difference Built-up Index)

- Pearson: 0.67
- Spearman: 0.61
- Interpretation: There is a strong positive correlation between NDBI and Land Surface Temperature. This implies that areas with more built-up surfaces (NDBI) tend to have higher Land Surface Temperatures. This aligns with the urban heat island effect, where artificial surfaces absorb and retain more heat (which strongly aligns with the hypothesis).

3. Median Household Income

- Pearson: -0.49
- Spearman: -0.48
- Interpretation: There is a moderate negative correlation between Median Household Income and Land Surface Temperature. This suggests that areas with higher median household incomes tend to have slightly lower land surface temperatures. This is potentially due to various factors, such as green spaces, tree cover, or better infrastructure in wealthier areas (such as more efficient HVAC systems, or newer, updated facilities).

4. Poverty Rate Percent

- Pearson: 0.45
- Spearman: 0.38
- Interpretation: There is a moderate positive correlation between Poverty Rate Percent and Land Surface Temperature. This indicates that areas with higher poverty rates tend to experience higher land surface temperatures. This often correlates with less green infrastructure, older housing areas, and more heat-absorbing surfaces in lower-income areas.

5. Renter Occupied Percent

- Pearson: 0.53
- Spearman: 0.47
- Interpretation: There is a moderate positive correlation between Renter Occupied Percent and Land Surface Temperature. This suggests that areas with a higher percentage of renter-occupied housing tend to have higher land surface temperatures. Similar to poverty rate, this might be linked to urban development patterns and access to cooling systems.

Overall Observations:

- NDVI and NDBI show the strongest correlations, showing the significant impact of vegetation and urban development on land surface temperature.
- The socioeconomic variables (Median Household Income, Poverty Rate Percent, Renter Occupied Percent) also show meaningful correlations, indicating a link between socioeconomic status and exposure to higher temperatures.
- The Pearson and Spearman correlation values are generally close for most variables, suggesting that the relationships are largely linear or at least consistently monotonic.

Interpretation of Kendall's Tau and Distance Correlations

Kendall's Tau

1. NDVI (Normalized Difference Vegetation Index)

- Kendall's Tau: -0.51
- This indicates a moderate negative monotonic relationship. As vegetation cover (NDVI) increases, Land Surface Temperature (LST) tends to decrease. This strengthens what we saw with Pearson and Spearman, reinforcing the idea that areas with more vegetation are generally cooler.

2. NDBI (Normalized Difference Built-up Index)

- Kendall's Tau: 0.44
- This shows a moderate positive monotonic relationship. As built-up areas (NDBI) increase, Land Surface Temperature tends to increase. This aligns with the urban heat island effect and is consistent with the other correlation types we previously saw.

3. Median Household Income

- Kendall's Tau: -0.33
- This suggests a weak to moderate negative monotonic relationship. As median household income increases, there's a slight tendency for Land Surface Temperature to decrease. While not a strong correlation, it shows a slight connection where more affluent areas might experience slightly lower temperatures.

4. Poverty Rate Percent

- Kendall's Tau: 0.26
- This indicates a weak to moderate positive monotonic relationship. As the poverty rate percentage increases, Land Surface Temperature tends to increase.

5. Renter Occupied Percent

- Kendall's Tau: 0.33
- This shows a weak to moderate positive monotonic relationship. As the percentage of renter-occupied housing increases, Land Surface Temperature tends to increase.

Distance

(Recall that moderate Distance Correlation values indicate significant dependency because any non-zero Distance Correlation shows that variables are not statistically independent, detecting relationships that previous methods might miss.)

1. NDVI (Normalized Difference Vegetation Index)

- Distance Correlation: 0.68
- A very strong distance correlation suggests a robust and significant dependency between NDVI and LST.

2. NDBI (Normalized Difference Built-up Index)

- Distance Correlation: 0.59

- A strong distance correlation indicates a significant dependency between NDBI and LST.

3. Median Household Income

- Distance Correlation: 0.45
- A moderate distance correlation indicates a notable dependency between median household income and LST.

4. Poverty Rate Percent

- Distance Correlation: 0.40
- A moderate distance correlation points to a significant dependency.

5. Renter Occupied Percent

- Distance Correlation: 0.46
- A moderate distance correlation indicates a significant dependency between renter occupancy and LST.

The Distance Correlation values show that even for variables where Pearson, Spearman, or Kendall's Tau show weaker associations, there is still a significant statistical dependency, meaning that none of these predictor variables can be considered independent of LST.

4.3 Heat Hotspot Identification

In the table below, we can see the mean and sample variance values of the environmental and socioeconomic data of each of the top ten and bottom ten census tracts as ranked by their land surface temperature.

Table 4.2: Mean and Variance

Category	Hottest Mean	Hottest Variance	Coolest Mean	Coolest Variance
Land Surface Temperature (C)	46.816	0.987	32.949	0.208
NDVI	0.302	0.008	0.743	0.001
NDBI	-0.066	0.001	-0.284	0.001
Median Household Income	37227.667	52318002.000	105557.500	199381078.500
Poverty Rate (%)	47.791	334.758	2.894	0.929
Renter Occupied (%)	91.914	28.376	10.872	17.621
65 Plus (%)	4.729	18.983	16.920	10.528
65 Plus Count	99.400	8368.711	882.100	85950.100
Elderly Living Alone (%)	43.489	599.115	19.262	52.999
Elderly Living Alone Count	48.500	2354.944	171.300	7435.567
Disability (%)	12.894	16.555	8.701	5.366
Disability Count	392.200	91356.400	445.700	20449.344
Nonwhite (%)	23.190	153.017	3.953	8.369
Nonwhite Count	655.600	92042.267	194.300	12570.011
Limited English Household (%)	1.629	6.176	0.634	0.589

Table 4.3 below shows the t-value of each category, calculated from the table above.

As we can see in the table, the most statistically significant categories excluding Land Surface Temperature, are NDVI, NDBI, Median Household Income, and Renter Occupied Percentage.

Table 4.3: t-values

Category	t-value
Land Surface Temperature (C)	40.107
NDVI	-14.770
NDBI	13.103
Median Household Income	-13.620
Poverty Rate (%)	7.749
Renter Occupied (%)	37.788
65 Plus (%)	-7.097
65 Plus Count	-8.059
Elderly Living Alone (%)	3.000
Elderly Living Alone Count	-3.925
Disability (%)	2.832
Disability Count	-0.506
Nonwhite (%)	4.789
Nonwhite Count	4.510
Limited English Household (%)	1.210
Limited English Household Count	0.567

NDVI and Median Household Income are negative, this means that both categories are higher on average in the coolest tracts. So the coolest tracts have more healthy vegetation on average while the hottest tracts have less, and the people living in the coolest tracts have a higher income on average while people living in the hottest tracts make less.

NDBI and Renter Occupied Percentage are positive, this means that both categories are higher on average in the hottest tracts. So the hottest tracts have more impervious surfaces such as concrete on average while the coolest tracts have less, and the people living in the hottest tracts are more likely to be renting on average while the people living in the coolest tracts are less likely to be renting. The t-value for the Renter Occupied Percentage is also very close to that of the Land Surface Temperature,

which tells us that the difference in Renter Occupied Percentage when comparing hotter and cooler tracts is massive. Looking at Table 4.2, we can see that the average Renter Occupied Percentage in the top ten hottest tracts is 91.9%. Comparing this with the average in the bottom ten coolest tracts being 10.9%, it is plain to see that renting is much more common in the hottest ten tracts of Lincoln than it is in the coolest.

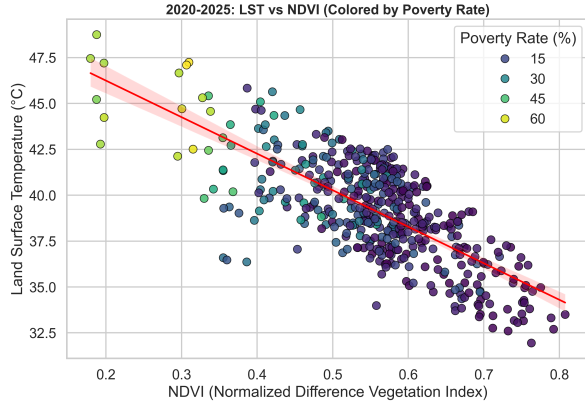
The most statistically insignificant categories are the Limited English Household Count and Percentage, and the Disability Count and Percentage categories. This means that these categories do not have a strong relationship with the Land Surface Temperature, and their values are independent of the Land Surface Temperature.

Once again, looking at [5], we can see that our hypothesis was correct in seeing a high correlation between land surface temperature and variables relating to income and poverty. Looking at [4], we can see similar results, which tells us that this method of correlating land surface temperature with other census tract variables is a viable method.

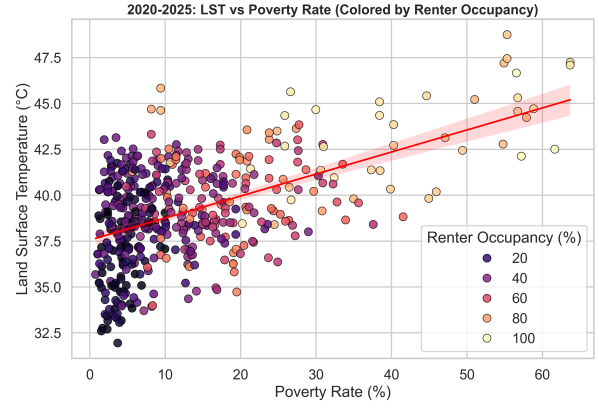
4.4 Data Integration and Visualization

Bivariate Relationships

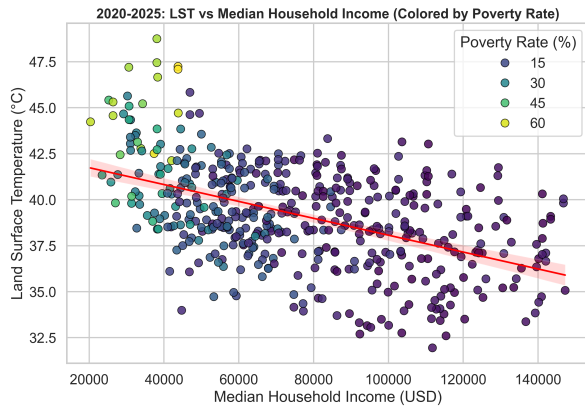
To quantify relationships between land surface temperature (LST) and key environmental and socioeconomic variables, we generated scatter plots with overlaid linear regression fits and confidence intervals. Each plot visualizes the bivariate relationship between LST and a predictor variable NDVI, poverty rate, median household income, and renter occupancy, while additional variables are incorporated through color scaling to provide additional context. The regression lines summarize the overall trend in each relationship and the confidence intervals indicate the uncertainty around these estimates. This



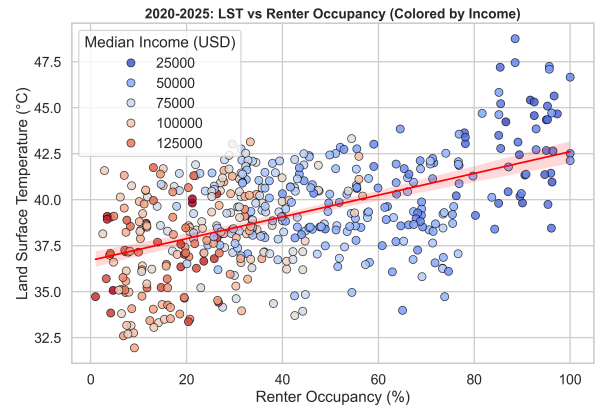
(a) Relationship between NDVI and land surface temperature (LST), colored by poverty rate.



(b) Relationship between poverty rate and LST, colored by renter occupancy.



(c) Relationship between median household income and LST, colored by poverty rate.



(d) Relationship between renter occupancy and LST, colored by median household income.

Figure 4.3: Relationships between land surface temperature (LST), vegetation cover (NDVI), and socioeconomic variables (poverty rate, median household income, and renter occupancy) for the period 2020–2025. Across all panels, LST is higher in areas with lower vegetation, higher poverty, lower income, and greater renter occupancy, highlighting spatial patterns consistent with urban heat inequality.

approach allows for both a visual and statistical assessment of how LST varies with vegetation cover and socioeconomic conditions, helping to identify consistent patterns and potential inequalities across the study area.

Figure 4.3a displays a higher vegetation cover (NDVI) with associated lower LST; areas with higher-poverty tend to cluster at lower NDVI and higher temperatures. Figure 4.3b shows LST generally increases with poverty rate, with higher renter occupancy areas concentrated at higher poverty levels and temperatures. Figure 4.3c indicates that areas with higher-income tend to exhibit lower LST, while regions with higher-poverty are associated with lower incomes and elevated temperatures. Figure 4.3d identifies areas with higher renter occupancy tend to have higher LST, while higher-income areas are generally associated with lower renter occupancy and cooler temperatures. Collectively, Figure 4.3 illustrate clear relationships between land surface temperature (LST), vegetation, and socioeconomic conditions over the 2020–2025 period. These results point to a consistent pattern of urban heat inequality, where disadvantaged communities, char-

acterized by lower vegetation, higher poverty, lower income, and higher prevalence are disproportionately affected by higher land surface temperatures.

4.5 Spatial Mapping and Clustering

4.5.1 Spatial Distribution of Heat

We began this report by examining the overall distribution of environmental and socioeconomic variables across census tracts in Lincoln, Nebraska, from 2020 to 2025. Spatial patterns of some of these variables are visualized in Figure 4.4. Higher temperatures are concentrated in central downtown, while vegetation (NDVI) is more prevalent in peripheral regions of Lincoln.

Urban Heat and Socioenvironmental Patterns in Lincoln, Nebraska (2020–2025)

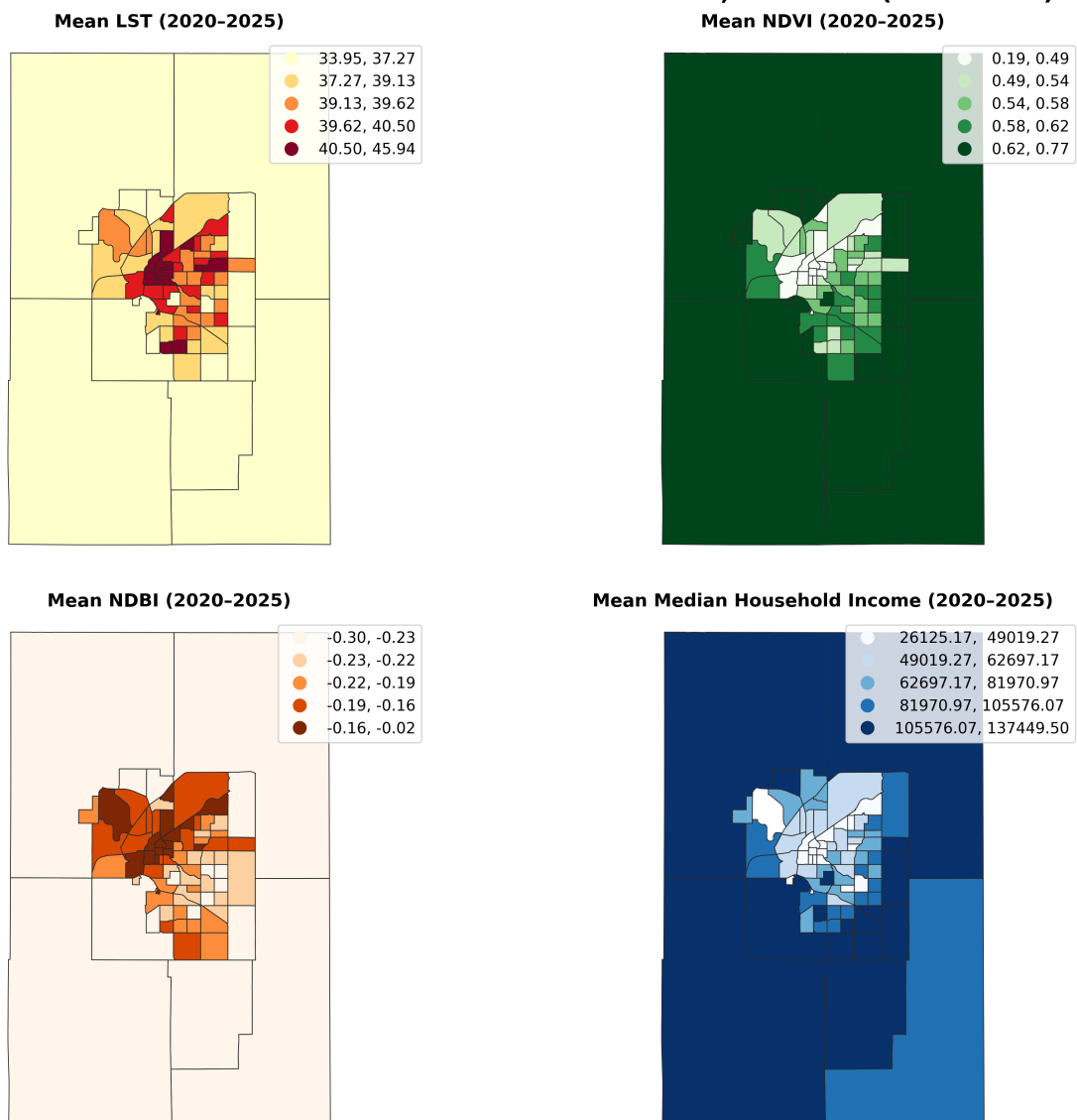


Figure 4.4: Spatial distribution of mean LST, NDVI, NDBI, and median household income (2020–2025)

These initial observations suggest a potential relationship between environmental conditions and socioeconomic characteristics.

4.5.2 Bivariate Spatial Relationships

To further examine the relationship between temperature and income, a bivariate choropleth map was constructed (Figure 4.5). This map classifies tracts into three quantiles for both LST and income.

Bivariate Map: Land Surface Temperature and Income (2020-2025)

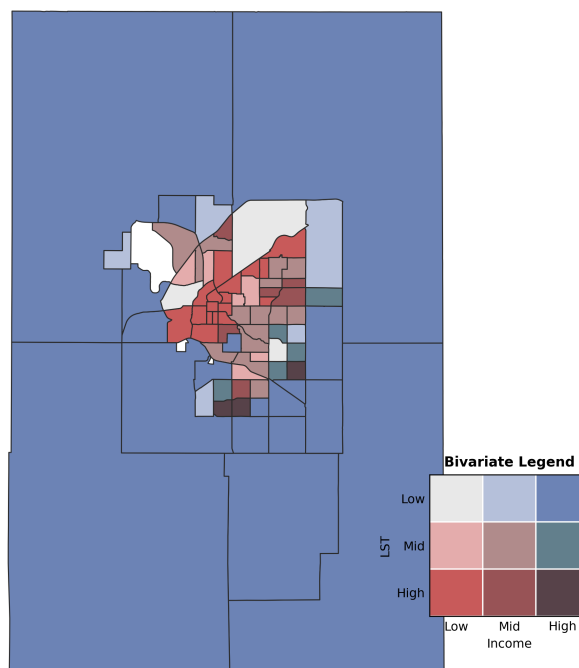


Figure 4.5: Bivariate map of land surface temperature and median household income

The distribution of tracts across categories is summarized in Table 4.4

Table 4.4: Bivariate Classification of LST and Income

Category	Number of Tracts
High LST – Low Income	17
High LST – High Income	3
Low LST – Low Income	3
Low LST – High Income	18

A substantial number of tracts (17) fall into the high temperature and low income category, indicating areas where environmental and socioeconomic vulnerabilities overlap. In contrast, 18 tracts exhibit low temperature and high income, reinforcing the presence of spatial inequality. This pattern suggests that heat exposure is not randomly distributed but is instead associated with socioeconomic status.

4.5.3 Spatial Autocorrelation Analysis

To formally assess spatial clustering, Global Moran's I was computed for LST. The results are shown in Table 4.5.

Table 4.5: Global Moran's I Results for Land Surface Temperature

Variable	Moran's I	Expected I	Z-score	p-value
LST	0.529	-0.0125	8.866	7.60×10^{-19}

The Moran's I value of 0.529 indicates a strong positive spatial autocorrelation, meaning that similar temperatures cluster geographically. The extremely small p-value confirms that this pattern is statistically significant and is unlikely to occur by pure chance.

To identify where clustering occurs, Local Moran's I (LISA) was computed and mapped (Figure 4.6).

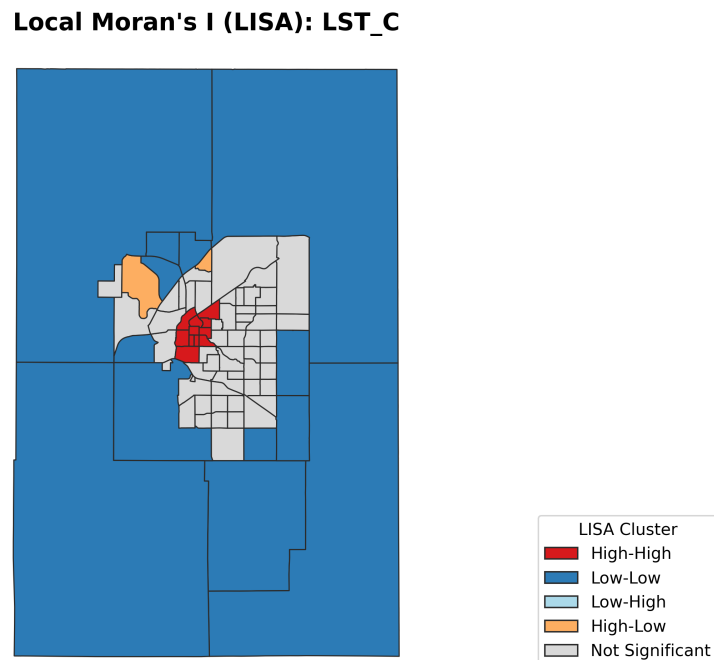


Figure 4.6: Local Moran's I (LISA) cluster map for LST

Cluster counts are summarized in Table 4.6.

Table 4.6: LISA Cluster Distribution for LST

Cluster Type	Number of Tracts
High-High	11
Low-Low	13
High-Low	2
Low-High	0
Not Significant	55

High–High clusters, representing hotspots, are concentrated in central Lincoln, while Low–Low clusters are located in peripheral areas. This confirms that urban heat is spatially clustered than evenly distributed.

4.5.4 Geographically Weighted Regression (GWR)

To examine spatially varying relationships between LST and explanatory variables, a Geographically Weighted Regression (GWR) model was implemented. The spatial distribution of R^2 values is shown in Figure 4.7.

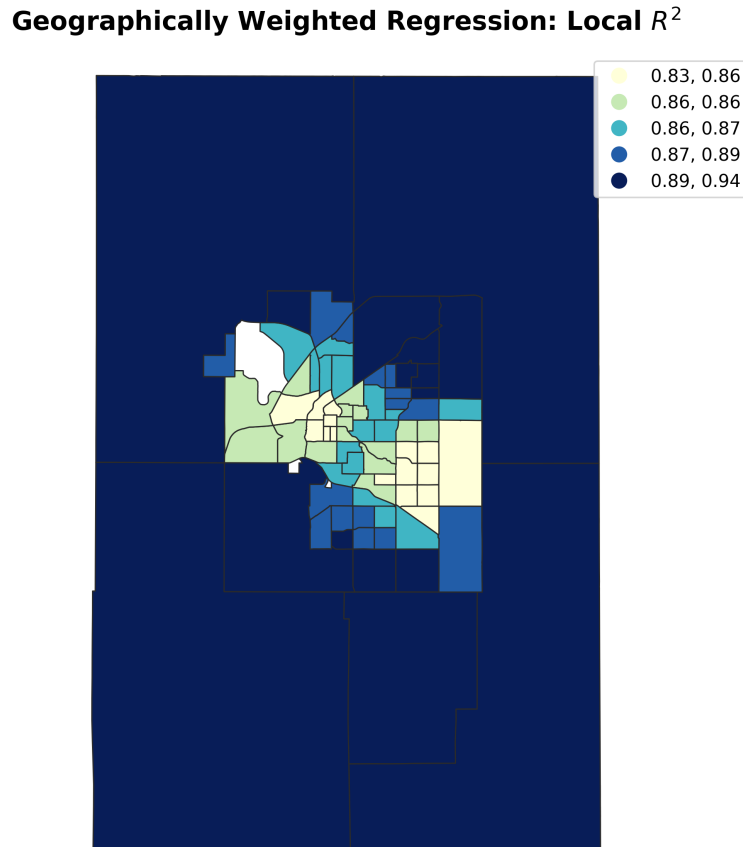


Figure 4.7: Local R^2 values from GWR model

Local R^2 values range from approximately 0.83 to 0.94, indicating strong model fit across most tracts. This suggests that the selected variables explain a big portion of variation in land surface temperature.

Figures 4.8–4.12 display the spatial variation in regression coefficients.

GWR Coefficient Map: median_household_income

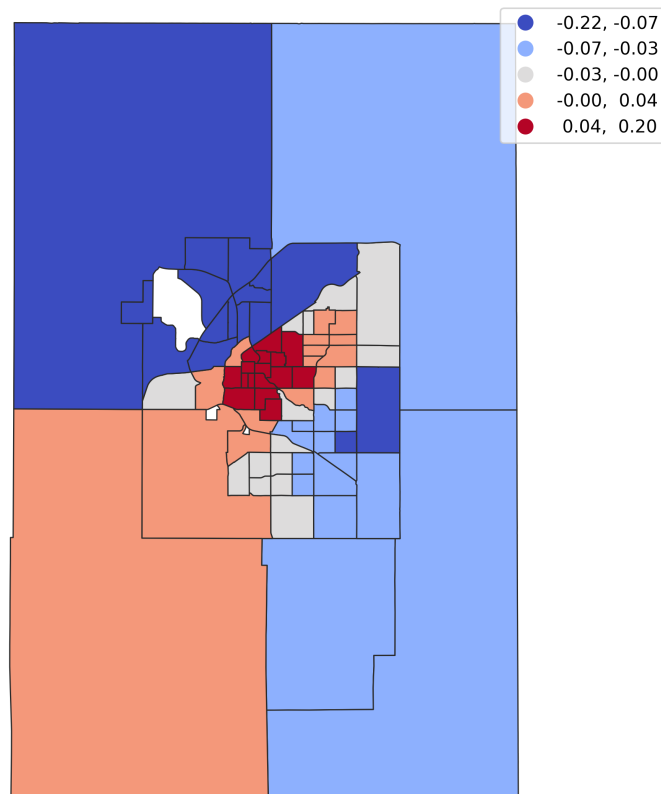


Figure 4.8: GWR coefficients for median household income

Median household income exhibits a negative relationship with LST in central tracts, indicating that higher-income areas tend to experience lower temperatures.

GWR Coefficient Map: NDVI

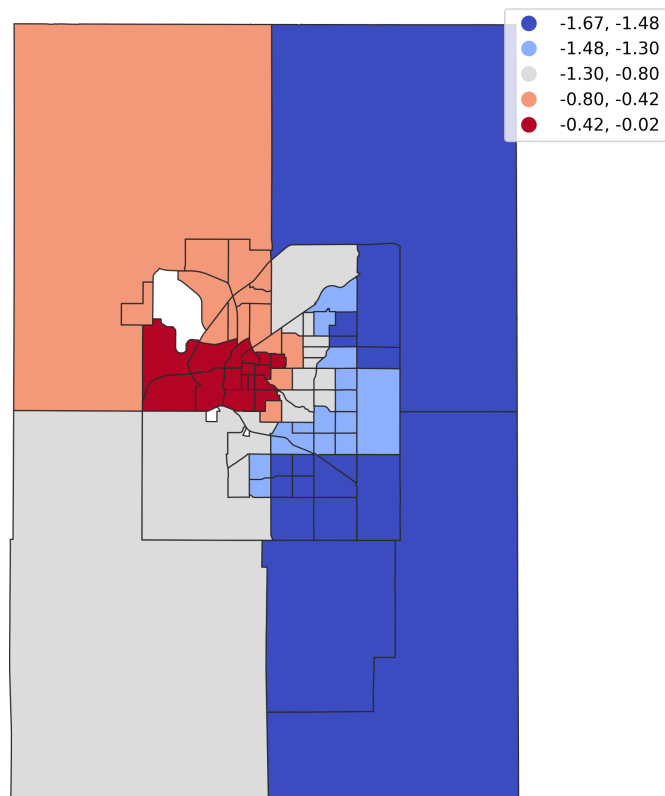


Figure 4.9: GWR coefficients for NDVI

NDVI shows a consistently negative association with temperature, reinforcing the cooling effect of vegetation.

GWR Coefficient Map: NDBI

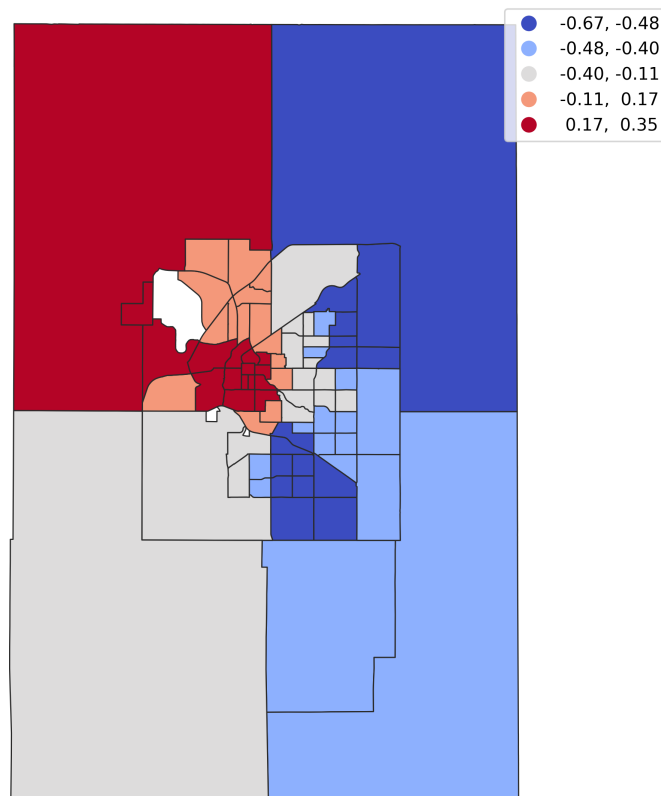


Figure 4.10: GWR coefficients for NDBI

NDBI demonstrates a positive relationship with temperature in many tracts, suggesting that built-up areas contribute to higher heat levels.

GWR Coefficient Map: poverty_rate_pct

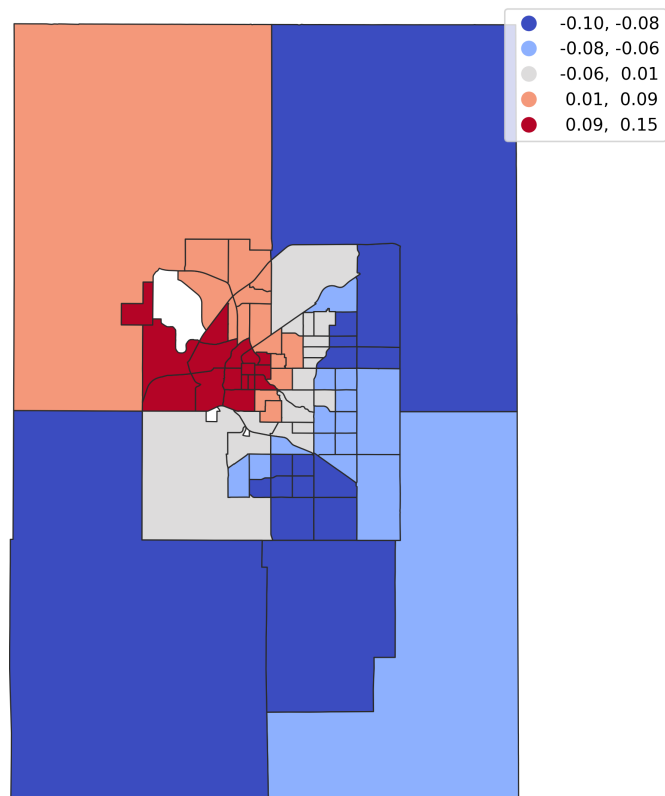


Figure 4.11: GWR coefficients for poverty rate

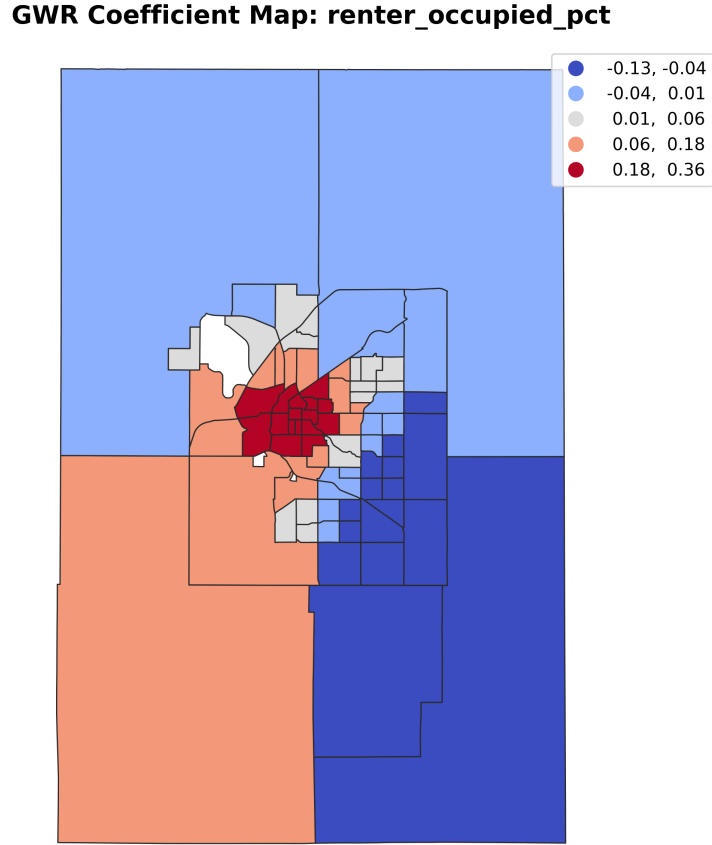


Figure 4.12: GWR coefficients for renter occupancy

Socioeconomic variables such as poverty rate and renter occupancy show spatially varying effects, indicating that their relationship with heat is context-dependent across different parts of the city.

4.5.5 Global and Spatially Varying Regression Results

To complement the spatial visualization and clustering analyses, both a global regression model and a GWR model were estimated to quantify the relationship between land surface temperature and explanatory variables.

Global Regression Results

Table 4.7: Global Regression Results

Variable	Coefficient	Std. Error	t-value	p-value
Intercept	0.000	0.048	0.000	1.000
NDVI	-1.139	0.139	-8.178	< 0.001
NDBI	-0.258	0.134	-1.934	0.053
Median Income	-0.046	0.091	-0.501	0.616
Poverty Rate	-0.078	0.087	-0.905	0.365
Renter Occupancy	0.046	0.098	0.466	0.641

The global model achieves an R^2 of 0.833, indicating that the selected variables explain a substantial portion of variation in land surface temperature. Among the variables, NDVI is the only statistically significant variable ($p < 0.001$), with a strong negative coefficient. This suggests that increased vegetation is associated with lower surface temperature.

NDBI shows a significant relationship ($p = 0.053$), indicating a possible association between built-up areas and temperature, though this is much weaker than the p value of vegetation. Socioeconomic variables, including median income, poverty rate, and renter occupancy are not statistically significant in the global model. This suggests that their relationship with temperature may not be constant across space.

Geographically Weighted Regression (GWR) Results

The GWR Model improves upon the global model by allowing relationships to vary spatially. Model information is summarized in Table 4.8.

Table 4.8: GWR Model Diagnostics

Metric	Value
Bandwidth	60.0
Residual Sum of Squares	6.844
AIC	66.223
Adjusted R^2	0.889

Compared to the global model, the GWR model achieves a higher R^2 of 0.912 and a lower AIC than that of the Global Regression (93.93), indicating improved model fit. This suggests that allowing coefficients to vary across space provides a more accurate representation of the factors driving temperature variation.

The spatial variability of coefficients is summarized in Table 4.9.

Table 4.9: Summary of GWR Coefficient Estimates

Variable	Mean	Std. Dev.	Min	Median	Max
Intercept	0.074	0.110	-0.273	0.083	0.276
NDVI	-0.979	0.525	-1.666	-1.154	-0.015
NDBI	-0.191	0.313	-0.668	-0.339	0.351
Median Income	-0.011	0.096	-0.218	-0.019	0.200
Poverty Rate	-0.007	0.082	-0.105	-0.043	0.153
Renter Occupancy	0.066	0.130	-0.133	0.031	0.361

4.5.6 Summary of Findings

The results of this section provide consistent evidence that urban heat in Lincoln, Nebraska is both spatially structured and socially differentiated. The mapping and summary statistics reveal substantial variation in land surface temperature across census tracts, with the highest temperatures concentrated in central downtown. This pattern is statistically confirmed by Global Moran's I, which indicates strong positive spatial autocorrelation, and further localized through LISA, which identifies a clear cluster of

high-temperature tracts in the downtown/center area. Environmental variables play a key role in explaining this pattern: vegetation (NDVI) is strongly and consistently associated with lower temperatures, while built-up areas (NDBI) generally correspond to higher temperatures. Additionally, bivariate mapping demonstrates that higher temperatures disproportionately occur in lower-income tracts, suggesting a spatial overlap between environmental and socioeconomic factors. While global regression results highlight vegetation as the dominant predictor of temperature, the GWR model reveals that these relationships vary across space, with improved model fit and heterogeneity in coefficient estimates. In conclusion, these findings indicate that urban heat islands in Lincoln are not randomly distributed[3], but instead reflects the combined influence of the environment and socioeconomic conditions, underscoring the need for spatially targeted approaches to heat mitigation and urban planning.

4.6 Machine Learning and Explainability

4.6.1 Random Forest and SHAP

In order to measure out-of-sample performance, we used Leave-One-Out Cross Validation (LOOCV), where each observation is predicted by a model trained on all other observations.

Model performance was:

$$\text{RMSE} = 1.799^{\circ}\text{C}, \quad \text{MAE} = 1.508^{\circ}\text{C}, \quad R^2 = 0.572.$$

These metrics indicate moderate predictive performance: the model explains approximately 57% of tract-level LST variance and produces average absolute errors near 1.5°C . The LOOCV R^2 reflects true out-of-sample generalization and can be seen as a conservative lower bound on model utility.

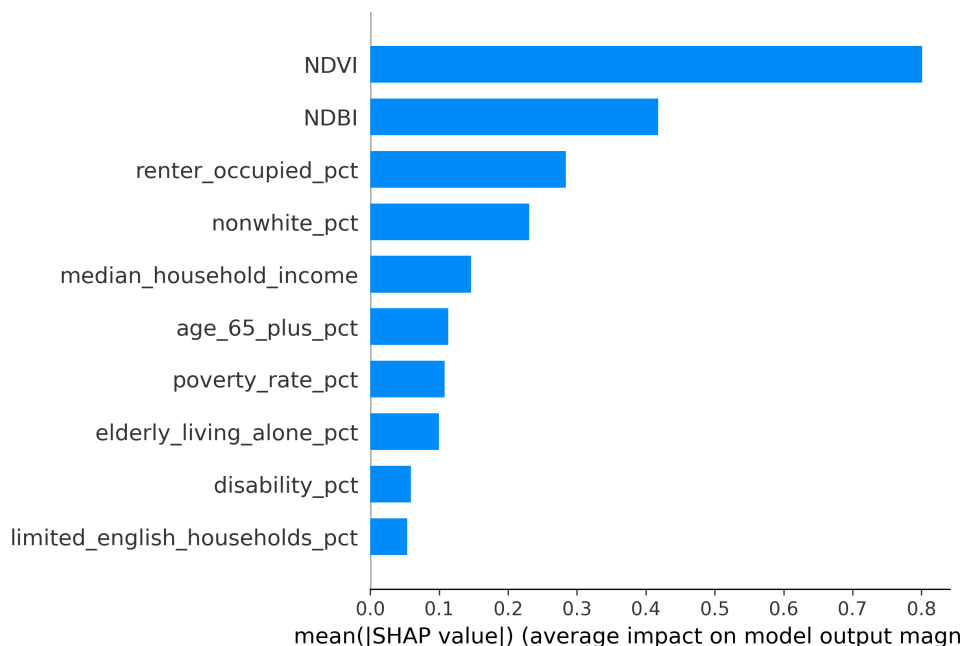


Figure 4.13: Mean absolute SHAP values by feature, ranked by importance.

The SHAP results reinforce the core UHI pattern: vegetation (NDVI) is the strongest cooling factor, while built intensity (NDBI) is the strongest warming factor. Socioeconomic structure remains important even after controlling for environmental variables, with renter concentration, poverty, and nonwhite percentage associated with higher predicted heat exposure.

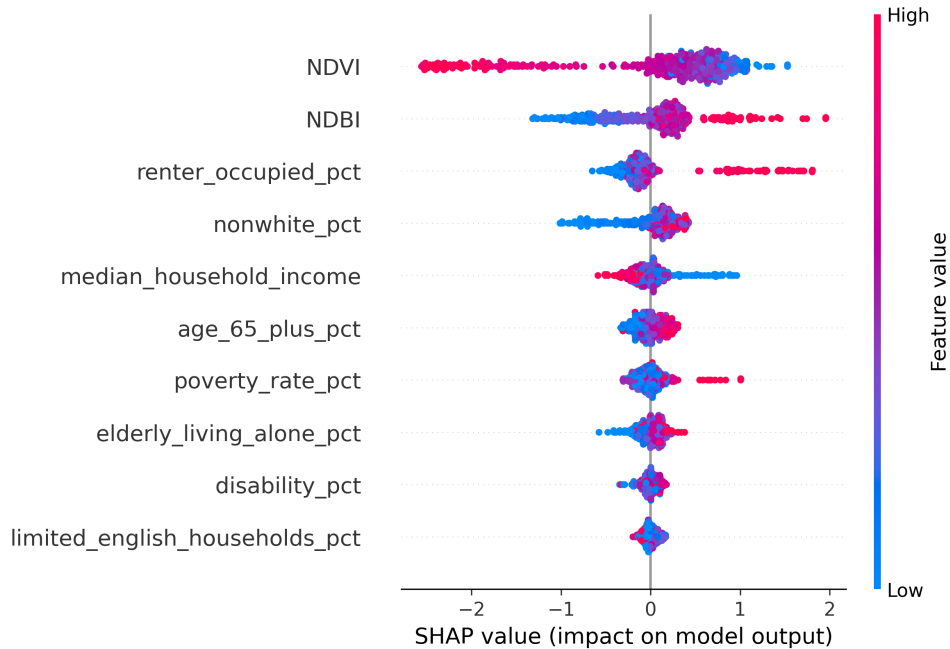


Figure 4.14: SHAP beeswarm plot. Each point is one observation and color indicates feature value

The beeswarm plot reveals directional effects of each feature, with color indicating the underlying feature value.

Environmental drivers: High NDVI tails leftward, while low NDVI clusters rightward, showing inverse correlation with LST as expected. NDBI shows the inverse pattern, with high built density driving strong warming with lower values correlating with reduced temperature.

Socioeconomic variables: Renter occupancy and non-white percentage show consistent positive association with LST independent of vegetation. Median income reverses: high-income trends cooler, low-income trends hotter, persisting after controlling for environment. Other socioeconomic variables contribute minimal predictive effect.

Chapter 5

Conclusions

5.1 Discussion

This study presents a novel, data-driven approach to understanding the Urban Heat Island effect in Lincoln, Nebraska, by integrating high-resolution satellite-derived environmental indices with detailed socioeconomic data at the census tract level. The longitudinal analysis spanning 2020 to 2026 offers unprecedented insight into how physical landscape features and social vulnerabilities intersect to shape local heat exposure patterns.

By adapting advanced spatial statistical methods such as Moran’s I and Local Indicators of Spatial Association (LISA), alongside interpretable machine learning techniques including Random Forest and SHAP analysis, this project not only identifies heat exposure hotspots but also elucidates the key environmental and demographic drivers behind these disparities. The use of zonal statistics to aggregate raster data to census tracts and the application of rigorous hypothesis testing (e.g., Welch’s t-test) further strengthen the robustness of the findings.

This work matters because it moves beyond descriptive analyses to actionable intelligence that can guide equitable urban planning and climate resilience strategies. The discovery that heat burden disproportionately affects low-income, renter-occupied, and minority neighborhoods in Lincoln underscores the urgency of targeted interventions.

Throughout this project, we developed and refined critical programming and computational skills, including satellite data processing with Google Earth Engine, spatial data aggregation using Python’s `rasterstats`, and the implementation of machine learning models with interpretability frameworks. These skills not only enabled the analysis but also provide a transferable toolkit for future urban environmental justice research. Ultimately, the findings empower local stakeholders with a scientifically rigorous foundation to prioritize investments in urban forest, cooling infrastructure, and community resilience, fostering a more just and sustainable urban environment for all Lincoln residents.

5.2 Recommendations for Lincoln’s City Council

The findings of this study demonstrate that urban dynamics and socioeconomic factors. Transitioning these insights into meaningful policy, these are the recommendations from this study for Lincoln’s City Council. The city should use the heat vulnerability maps generated in this study to prioritize the installation of tree canopies and cool pavements

in high risk census tracts. Targeted outreach programs should be developed for residents with limited English proficiency and those living alone, ensuring that life-saving resources reach those facing the greatest barriers. The council should implement localized environmental monitoring in significant non-residential areas, such as the Lincoln Airport, where data gaps currently prevent a full understanding of the thermal risks faced by workers. Using satellite-derived surface temperature as a key performance indicator for the 2021 Lincoln Climate Action Plan, ensuring that urban development remains aligned with heat mitigation goals. Linking heat vulnerability data with specific health outcomes, such as heat-related illnesses, the city can allocate emergency services and design long term strategies.

5.3 Limitations

Although the study is an assessment of the spatiotemporal dynamics of Lincoln’s UHI effect, several limitations were tied with the data and methodology and provide opportunities for further refinement.

5.3.1 Temporal Constraints

The analysis was primarily restricted to a five-year window (2020-2025). The absence of a robust pre-2020 baseline limits the ability to quantify how the COVID-19 pandemic and shifts in urban activity altered thermal patterns. This restriction makes it difficult to distinguish long term climatological trends from short term anomalies.

5.3.2 Data Gaps in Non-Residential Tracts

Certain census tracts within the study area lacked socioeconomic and demographic data. These gaps were most prevalent in non-residential areas, such as areas occupied by the Lincoln Airport or correctional facilities, where traditional census metrics do not accurately reflect the daily population or usage of the land.

5.3.3 Transportation

The current model does not fully capture recent changes in transportation behavior, specifically the increased reliance on personal vehicles over public transit. Since transportation emissions and infrastructure are significant contributors to anthropogenic heat, omission may result in an underestimation of heat intensity in high traffic locations.

5.4 Future Work

After fully reviewing the analysis, potential future research would be essential to mitigating future climate implications. Future studies should extend beyond 2025. This will be essential for evaluating long term efficiency of the Lincoln 2021 Climate Action Plan and determining if policy interventions are successfully mitigating heat inequality. Natural progression of this study involves correlating the developed heat vulnerability maps with public health datasets. Analyzing heat-related hospitalizations and mortality rates at the tract level would provide a more direct understanding of how thermal inequity impacts

community wellness. Moving from census-tract-level analysis to census block groups or parcel-level data would provide a microscale granularity required by urban developers and city planners to implement targeted cooling interventions, such as cool pavements. Future initiatives should focus on the human element of heat safety. This includes developing tailored outreach strategies for Lincoln's most vulnerable populations, specifically residents with limited English proficiency or those living alone, to ensure equal access to cooling centers and emergency resources.

5.5 AI Use Statement

AI Policy: AI may support your learning by helping to clarify concepts, reveal connections, and improve efficiency. You are responsible for all submitted work and must be able to explain and justify each key step.

Peck, Bryant

Similar to others, I utilized AI to refine my code. For the most part, this was debugging and finding slight errors throughout, which was more efficient than having to take time to do this myself. This left more time for analysis. My main contribution was the correlation analysis, and inputting the formulas into Python was sometimes difficult. AI assisted me in correcting the mistakes I had so the output displayed the way I intended.

Martinez, Helen

For my section, I used AI to find any logical holes in my analysis and the mathematical background that could potentially raise any red flags. Additionally, I used AI to help create the figures. In our preliminary analysis, I used a program called QGIS, but it was very time-consuming considering the number of figures we have for this final report. Therefore, I switched to Python and consulted AI to give me starter code to create the visuals using Python's Matplotlib.

Pettis, Josie

For my sections, I utilized AI to refine the code for the bivariate relationship scatter plots to make them cleaner once the 2020-2025 data were plotted, such as formatting elements like axis scaling and legend placement to enhance readability. AI also helped me clarify any connections that I missed between data constraints and helped generate ideas for the Lincoln City Council. Selecting the research variables were decided within our group including Landsat and Census data.

Pandit, Sahara

For my section, I used AI to help with the code that created the histograms, some of the histograms weren't showing up or were inaccurate, so it helped me distinguish the independent variables in each graph and make sure that everything was showing up accurately.

Waller, Thomas

For my section, I used AI to help with coding and plot making in python. I also used it to help with model parameter values and validation type, and

it helped explain the reasoning behind proposing them, which I confirmed through outside sources. I also used AI to check my work, such as verifying code and correcting grammar or word choice in the results interpretation.

Hayden, Xavier

For my part, I used AI to help compare and contrast different mathematical methods so that I could make a more informed decision on what mathematical route to take. I also used it to make a cursory check that my work was correct, before looking through it fully again. I did not use it to generate text or images to be used in any part of this project.

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